

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Consequence Machine: Interactive S
<i>Company:</i>	Please provide company name

Contact Details

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Additional Info

<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Human-Computer Interaction; Responsible AI, Critical and Speculative Design; Technology and Society

Project Background Information

The technologies we design can create consequences that are difficult to anticipate during the design process. Real-world examples abound of unintended consequences and wider societal impacts resulting from technology deployment. However, these cases are typically documented after the fact and are rarely connected to decisions made during design. Research has shown that transforming real-world ethical challenges into interactive scenarios can make these issues more salient and engaging for designers.

In this project, you will design and develop a web interface that takes a user's design idea as input and identifies relevant, documented real-world cases. You will build a system that extracts the key consequences, events, and lessons from these cases and uses them to construct an interactive scenario around the user's proposed design. The scenario should allow users to explore how their design might play out in the real world. Importantly, users should also be able to trace consequences they encounter back to the original real-world cases that informed the scenario.

¹ An overview of different research areas in the school of computer science can be found [here](#).

The main goal is to encourage more critical thinking about technology design by helping designers question their assumptions, consider potential consequences, and explore alternative design choices.

Requirements

The group must provide:

Web application: a functional web interface where users can submit a design idea and interactive with a generated scenario.

Scenario generation system: running in the backend of the web app is a system that retrieves real-world cases from the web relevant to the user's design idea and use them to construct an interactive scenario. The system will leverage LLM, where appropriate, to synthesise information from these cases into relevant events, decisions, and consequences that users can interact with and explore.

User evaluation: a study evaluating the usability, engagement, and effectiveness of the experience.

Code and documentation: documentation of system architecture, implementation, and evaluation results.

Skills

Highly desirable:

- Web development (frontend and backend)
- Curiosity and critical thinking
- Ability to work in a team

Desirable:

- LLM/generative AI development
- UI/UX design

Submission / Award information

<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.